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Ollscoil Teicneolaíochta na Mumhan
Munster Technological University

SESSION 1 · FRIDAY 18 SEPTEMBER 2026

Understanding AI: Opportunity, Limitations and Trust

Building AI Confidence in the Gaeltacht Workplace

Údarás na Gaeltachta × Munster Technological University

Dr Haithem Afli

10:00–15:30 · Campas Íosagáin

Meet Aoife.



Aoife starts her day.

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Her workload grows.



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Emails • funding proposal • survey feedback • newsletter • Irish communication.

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She looks at the workload.



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Could AI actually help?

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Welcome and expectations

Five minutes, then straight to Mary. This is not a technical AI course.

- Purpose of today: build confidence to make good decisions about AI, not to become an expert.
- Everyone here has a different level of AI experience — that is expected, not a problem.
- Scepticism is welcome. Nobody needs to be convinced of anything today.
- Structure: morning to understand, afternoon to try — two workshops, with time to reflect after each.
- Laptops/phones stay closed this morning; you will want them for the afternoon workshops.
- Coffee break mid-morning and mid-afternoon, lunch break at 12:25, finishing at 15:30.
- Ground rules: be curious, be sceptical, protect information, check AI outputs, keep humans responsible.

Welcome, then Mary — morning to understand, afternoon to try. Two workshops, two reflections.

MORNING · understand

- 10:00** Welcome and expectations
- 10:05** Cybersecurity foundations · Mary
- 10:25** What do you think? (the concerns wall)
- 10:35** AI in the Irish context
- 10:55** What actually is AI?
- 11:15** Coffee break
- 11:30** Software, automation or AI?
- 11:55** Large language models
- 12:15** Live demonstration

AFTERNOON · try

- 12:25** Lunch break
- 13:10** What can AI do for my work? · Five live demos
- 13:30** Real case: could AI get this wrong? · GAA
- 13:40** Workshop 1 · Can AI help me? · 5 groups of 3
- 14:05** Reflection 1 · What did we learn?
- 14:15** Workshop 2 · Prompt challenge
- 14:40** Coffee break
- 14:55** Reflection 2 · What changed?
- 15:05** My first AI experiment
- 15:20** What we said today · Close

From protecting ourselves to inviting AI in

Mary has just taken us through how we protect ourselves and our organisation from digital threats.

Now let us look at what happens when we actively invite an AI system into our work.



A phishing email

Something is trying to get in without being asked. The instinct is to keep it out.



An AI tool

We are choosing to hand something over. The instinct has to be different, and more deliberate.



Same care, new question

Not “is this safe to open?” but “what am I about to give away, and what will I do with what comes back?”

Today's question

What is AI, why does it matter to my work, and why should I not blindly trust it?

THE JOURNEY THROUGH TODAY

What is AI?



What can it do?



What can go wrong?



How can I use it?



How can I use it safely?

You do not need to become an AI expert. You need to become confident enough to make good decisions about AI.

Before we start: what do you think?

Three questions. There are no wrong answers, and nothing anyone says today will be argued with.



First thought

When you hear “Artificial Intelligence”, what is the first thing that comes into your mind?



Who has used one?

Hands up if you have used an AI tool, for anything, even once.



Who has avoided one?

Hands up if you have deliberately decided not to. That is a legitimate position and it is welcome here.

Every concern goes on the wall. We come back to it at the end of today, and again in Sessions 2, 3 and 4.

What you told us

Filled in live. These are the questions the rest of the programme has to answer.

Trust and accuracy

...

Answered: Today, and again in Session 2

Privacy and confidentiality

...

Answered: Today in outline, fully in Session 2

Work and jobs

...

Answered: Today, revisited in Session 4

Irish language

...

Answered: Touched today, opened in Session 3

Rules and permission

...

Answered: Session 2

Something else

...

Answered: We will find the right place for it

All three are true

AI is exciting, useful and worrying

Most rooms split into these three. All three positions are reasonable, and all three are partly right.



Exciting

It does things that were not possible three years ago, and it does them in seconds.



Useful

For a small organisation with more to do than hands to do it, that matters more than novelty.



Worrying

It is confident when it is wrong, it is hungry for information, and nobody asked us before it arrived.

Today is not about deciding which of these is correct. It is about being able to hold all three at once.

Cybersecurity foundations

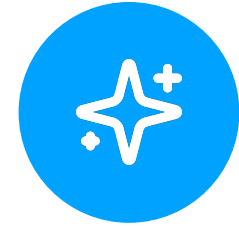
Delivered by Mary. Protecting ourselves and our organisation online — the material for this section lives in her own session, not in this deck.

What are we allowing into our organisation?

The question this section answers. In twenty minutes, we turn it around and ask the AI version of the same question.

ACT 2 • 10:35 • AOIFE'S FIRST QUESTION

What is AI?



“Wait — what actually IS the thing I keep hearing about?”

Already around us



In Ireland



And Irish



Not one thing



Software → ML → generative

AI is already around us

Before any definition: there is a good chance that most people in this room used an AI-enabled system before they arrived this morning.



Predictive text



Spam filtering



Voice assistants



Maps and traffic



Recommendations



Search



Translation



Photo and camera



Banking fraud alerts



Document scanning



Marketing tools



Customer chat

None of this was announced. It arrived quietly, and most of it is useful.

Where is AI already being used in Ireland?

Ask the room before showing this. The answers are usually better than the slide.



Public services

Application processing,
correspondence triage, translation of
public information.



Business

Marketing copy, customer service,
forecasting, stock, recruitment
screening.



Tourism and culture

Listings, itineraries, multilingual visitor
information, transcription of
recordings.



Everyday admin

Meeting notes, summaries, drafting,
scheduling, first-pass document
review.

And the harder question: where might AI already be affecting Irish-language communication?

Can AI understand Irish?

Does Irish-looking AI output mean good Irish? Pause on that. The honest answer, given now, so nobody leaves today with the wrong impression.

WHAT IS TRUE

- It can produce Irish, often fluently.
- It is markedly better at English than at Irish.
- Fluent Irish and publishable Irish are not the same thing.
- The gap is about the amount of digital Irish it learned from, not about the language itself.

WHAT WE WILL NOT CLAIM

- That it is “good enough” in Irish. That depends entirely on the use.
- That it is useless in Irish. It is not.
- Any number, benchmark or percentage. We will not invent one.

AI for Irish — not AI instead of Irish.

Why Irish matters

About two minutes — not an academic talk.



MY RESEARCH

- My research is about language and AI, particularly multilingual and low-resource languages.
- Irish is a really interesting example because AI systems are much stronger in some languages than others.

THE QUESTION I'M ASKING

- So today I'm not only interested in what AI can do.
- I'm interested in asking: can we use AI in a way that strengthens Irish rather than simply asking Irish speakers to adapt to systems built primarily for English?

Where could poor Irish AI output matter in your own work?

AI is not one single technology

It is an umbrella over a set of quite different things, which is why general claims about “AI” are usually wrong.

Machine learning

Computer vision

Speech recognition

Recommendation systems

Natural language processing

Generative AI

Language models

AI ≠ ChatGPT. ChatGPT, Claude, Gemini and Copilot are products built around models. The product is not the technology.

Confusing the two is how people end up believing that “AI cannot do maths” or “AI is always out of date”. Those are facts about one product on one day.

Traditional software

A person wrote every rule. The machine applies them exactly.

Rules



Output

WHAT THAT MEANS

- Someone decided, in advance, what should happen in every case.
- Give it the same input twice and you get the same answer twice.
- It is right because the rules are right, not because it learned anything.
- When it is wrong, you can find the line that is wrong and fix it.



WHERE YOU MEET IT

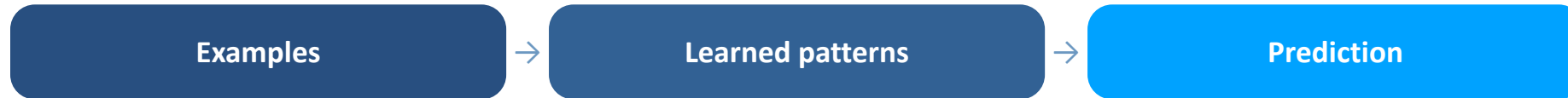
Your calculator, your payroll system, the formula in your spreadsheet.

Quick hands-up: is your calculator “AI”?

A. Yes · B. No · C. It depends

Machine learning

Nobody wrote the rules. The system worked them out from examples.



WHAT THAT MEANS

- You show it thousands of cases and it infers what they have in common.
- It produces a prediction, often with an associated confidence or probability.
- Its behaviour can change when the data, training or model changes.
- When it is wrong, there is often no single line to point at.

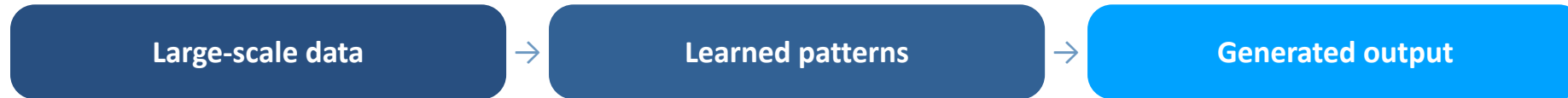


WHERE YOU MEET IT

Fraud detection, traffic prediction, spam filtering, recommendations.

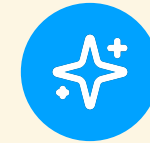
Generative AI

It produces something that did not exist before.



WHAT THAT MEANS

- Trained on very large amounts of text, images, audio or code.
- It generates new output rather than simply selecting from a fixed list.
- The output is fluent by design, which is exactly what makes it risky.
- Two identical questions can produce two different answers.



WHERE YOU MEET IT

ChatGPT, Claude, Gemini, Copilot, image and voice generators.

Thirty seconds, pair up: which of these have you used this week?

ChatGPT · Copilot · Gemini · None of them

Coffee break

11:15 – 11:30 · Fifteen minutes

When we come back: is a spell checker AI?

Software, automation or AI?

Six examples. Sort each one. Two of them will split the room, and that split is the lesson.

Traditional software



Automation



AI



Generative AI

1 A calculator working out €200 + €350

2 An email rule moving “invoice” messages to a folder

3 A fraud system learning unusual transaction patterns

4 An AI tool writing this month’s newsletter

5 Google Maps predicting traffic on the way to Galway

6 The spell checker in your word processor

And why the disagreement is the point

1	Calculator	Traditional software	Rules a person wrote, applied exactly. Right because arithmetic is right.
2	Email rule	Automation	A person set the condition and the action. Nothing was learned. It saves work, which is why people call it AI.
3	Fraud detection	AI	Nobody wrote the rule for “unusual”. It was inferred from examples and it keeps changing.
4	AI writing the newsletter	Generative AI	It produces text that did not exist before. The output is new; the reliability is your problem.
5	Maps traffic	AI	Prediction from learned patterns. Not generative: it is forecasting, not writing.
6	Spell checker	It depends	A dictionary lookup is traditional software. Modern predictive suggestions are machine learning. Same menu item, different technology.

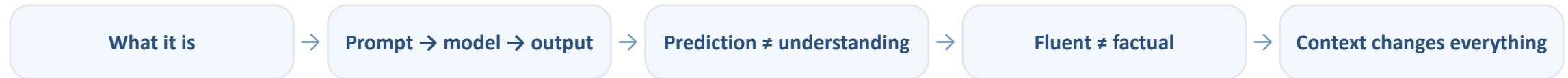
Do not accept “AI” as a magic label. Understand what the system actually does.

ACT 2 CONTINUED • 11:55 • ONE LEVEL DEEPER

Large language models

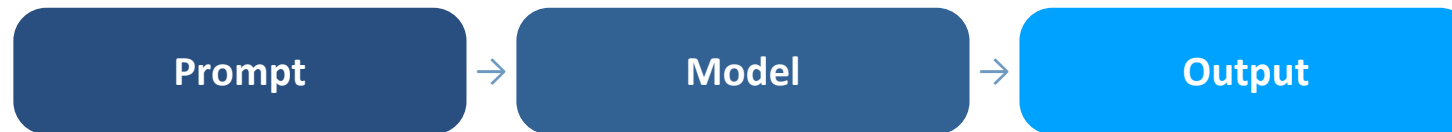


Nobody has ever actually explained this to Aoife.



What is a large language model?

A large language model learns statistical patterns from very large amounts of language, and uses those patterns to generate a response to what we give it.



Context

What we put in shapes what comes out. Almost everything you can control is here.



Probability

At its core it generates likely continuations from learned patterns. Some AI products can also connect to outside information and tools.



Generation

The answer is composed, word by word. It did not exist before you asked.

Guess before we say it: has this model seen more words than...

A. A library · B. A country's newspapers, ever · C. Most of the public internet

Prediction is not understanding

It can produce a perfect paragraph about something it has no concept of.

WHAT IT DOES NOT DO ON ITS OWN

- By itself it does not automatically know whether what it has generated is true.
- It does not reliably tell the difference between a fact it learned and a plausible sentence it composed.
- It has no memory of your organisation, and no reliable internal sense of “I am not sure about this”.

WHY THAT MATTERS AT WORK

- You cannot delegate judgement to it, only drafting.
- A confident tone tells you nothing about accuracy.
- The check comes from outside the system. From you, or from a source you can point at.

Would you send Aoife’s draft to a customer without reading it first?

Yes · No · Depends what it says

Fluent is not the same as factual

The tone of an answer tells you nothing about whether it is true. This is the single hardest habit to build.

“Our seaweed products are certified organic and have won three national food awards.”

FICTIONAL EXAMPLE. Precise, confident and entirely invented. Precise details can sound authoritative even when they are invented.



Numbers

Percentages, amounts and dates are where fabrication concentrates. They look like the shape of a real answer.



Names

Scheme names, report titles, citations and legislation. Perfect formatting is not evidence of existence.



Quotations

Anything in quotation marks needs finding at source before it goes anywhere near print.

The prompt is the variable

Same question. Three audiences. Run live, one after another, without tidying the results.

To a seven-year-old

Explain what a large language model is to a seven-year-old.

To a small business owner

Explain what a large language model is to a small business owner.

To a computer science lecturer

Explain what a large language model is to a computer science lecturer.

What changed?

The prompt changes the response. Context matters.

Lunch break

12:25 – 13:10

This afternoon: you use it. Bring a laptop or a phone back with you.

ACT 3 · 13:10 · AOIFE'S AFTERNOON

What can AI do for my work?



“OK. Let’s actually try it on my real list.”



AI as an assistant, not an authority

A capable, fast, slightly unreliable colleague who has never met your customers and will not tell you when they are guessing.



DELEGATE

Good at

First drafts, alternatives, summaries, structure, boring scaffolding, second opinions.



SUPERVISE

Needs checking

Anything factual, anything in Irish, anything with a number, anything going out under your name.



KEEP

Never

Decisions about people, sensitive communication, final approval, and accountability — and anything Mary's section told you never leaves the building.

You remain responsible for what you use and publish.

What never goes into a public AI tool

The same instinct from Mary's cybersecurity section applies here — what are we handing over, and to whom?

DO NOT ENTER

- Confidential information
- Personal information
- Customer information
- Commercially sensitive information
- Passwords or credentials
- Private organisational information
- Sensitive documents
- Anything your organisation's policy says must not leave the building

Use fictional, public or anonymised examples in every workshop this afternoon.

Generate content

Aoife needs a Saturday-market post for a seaweed producer — in twenty minutes, not twenty prompts.

Create a short promotional post for a fictional Gaeltacht food producer selling seaweed products at a Saturday market.

Make it more conversational. Shorter sentences. No exclamation marks.

Now give me an Irish version and a natural English version. Do not translate the Irish word for word: write the English properly. Then list any Irish terms you were unsure about.

WHAT TO POINT AT

How fast the register can be steered, and how the bilingual request changes the shape of the work.

SHOW THE WEAKNESS TOO

It invents detail it was never given: a price, an opening time, an award. Point at the invented detail on screen.

Aoife's decision: usable as a starting point — she still checks every invented detail before it goes out under the business's name.



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Before we run it: what tone do you expect the first draft to have?



Example output (illustrative)

"Fresh Atlantic seaweed, hand-harvested this week. Available at the Saturday market from 9am."

Edit a proposal

A funding proposal needs a second pair of eyes and Aoife has nobody free until Thursday.

Here is a draft funding proposal for a fictional community arts project. Improve the structure without changing what it claims. Tell me what you moved and why.

Which sentences would a reader outside this sector find unclear? Quote them.

What three questions would an assessor most likely ask about this proposal? What is missing?

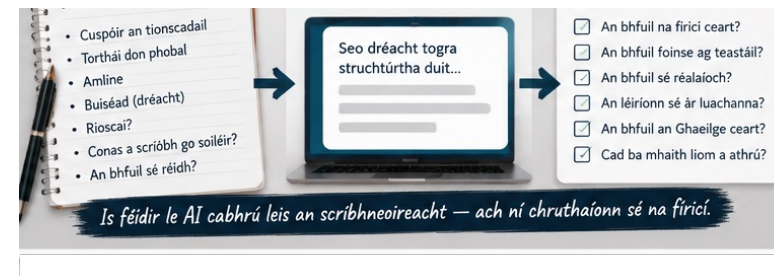
WHAT TO POINT AT

The reviewer-questions prompt is the valuable one. It finds gaps the author cannot see in their own text.

SHOW THE WEAKNESS TOO

Ask it to improve clarity and it will quietly change meaning. Diff the claim, not just the prose.

Before we run it: will the model change what the proposal claims, or just how it reads?



Aoife's decision: the reviewer-questions prompt earns its place — the rewrite prompt gets diffed against the original, not trusted on sight.

Create a newsletter

The monthly newsletter is overdue, in two languages, and Aoife has forty minutes before the school run.

Write a short monthly newsletter for a fictional Gaeltacht community organisation. Open evening, last Thursday, 7pm, community hall. Audience: local members who already know us. Also: a new craft group starting, and the AGM date to be

Now give me a version half that length for people reading on a phone.

Five subject lines. None of them clickbait.

Now the Irish version, written as Irish rather than translated. Flag anything you are unsure about.

WHAT TO POINT AT

Four useful outputs from one set of facts, in about ninety seconds.

SHOW THE WEAKNESS TOO

Watch what it does with “AGM date to be confirmed”. Models dislike loose ends and often invent a date.



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Before we run it: what will it do with the one detail we left vague — the AGM date?



Example output (illustrative)

"Open evening — last Thursday, 7pm, community hall. New craft group starting. AGM: [date TBC]"

Aoife's decision: four drafts in ninety seconds — but the loose end (“AGM date to be confirmed”) is exactly what she checks first.

Analyse survey feedback

Forty feedback forms sit in a folder. Aoife needs the real themes, not a comforting summary.

Here are 40 anonymous responses from a fictional training feedback form. What are the three strongest themes?

What are participants unhappy about?

What actions would you recommend?

Now show me the exact responses that support each conclusion. Quote them. If a theme is supported by fewer than three responses, say so.

Before we run it: do you expect the summary to name its evidence, or just assert it?



Example output (illustrative)

*"Theme: venue access (6 mentions).
Theme: pace (11 mentions).
Theme: catering (2 mentions)."*

WHAT TO POINT AT

The first three prompts give a confident summary. The fourth is what makes it trustworthy, and sometimes dismantles it.

SHOW THE WEAKNESS TOO

A theme can be real and still be two people out of forty. Without the evidence prompt you cannot tell.

Aoife's decision: the fourth prompt — asking it to quote the evidence — is what turns a confident summary into a trustworthy one.

Ask questions of documents

A funding scheme's terms run to eleven pages. Aoife needs one straight answer before 5pm.

Using only the attached documents, what does this scheme cover and what does it exclude? Quote the passage you used for each answer.

What does the deadline appear to be? If the documents do not say, tell me they do not say.

WHAT TO POINT AT

Give it the documents and demand quotations and you get something much closer to verifiable.

SHOW THE WEAKNESS TOO

The second prompt is the real test. A model that invents a deadline rather than admitting the documents are silent has told you how much to trust the rest.

Before we run it: what should it say if the documents simply don't cover the question?



Example output (illustrative)

*"The documents do not state a deadline for this scheme.
(No matching passage found.)"*

Aoife's decision: a model that admits the documents are silent has just told her how far she can trust the rest of its answers.

Are @officialgaa using AI art now?



Barry Masterson 🏈
@BarryMasterson · Follow



So are @officialgaa using AI art now? There are a ton of Irish artists who specialise in GAA illustration you could commission and support.

Things are tough at the best of times but I might have to pack it in at this rate honestly if this is what we have to look forward to 🙄



7:42 AM · May 18, 2024



❤️ 1.5K 💬 Reply 🔗 Copy link to post

Read 98 replies

SOURCE

Irish Independent, "GAA defends use of AI applications to create artwork for match programmes following criticism", 18 May 2024.

independent.ie/irish-news/gaa-defends-use-of-ai-applications-to-create-artwork-for-match-programmes-following-criticism/a/129643374.html

What's wrong here?

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But is the problem only the strange hands and helmets?



Zoomed detail, real artwork: the county name is misspelt — “CIR COGHAIN” for Tir Eoghain (Tyrone).

CONTEXT

CULTURE

COMMUNITY

EXPERTISE

HUMAN JUDGEMENT

AI can generate something that looks culturally recognisable without necessarily having sufficient cultural or contextual judgement.

There are Irish artists who specialise in GAA illustration — this is also a livelihood question, not only an accuracy one.

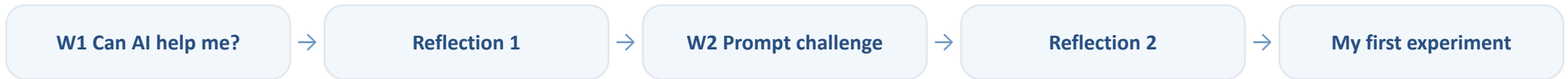
AI can be culturally recognisable without being culturally appropriate.

ACT 5 • 13:40 • NOW YOU HELP AOIFE

Afternoon workshops



Two workshops, two reflections. 5 groups of 3. You make the calls Aoife would have to make.



Can AI help me?

5 groups of 3: task owner, AI operator, checker — roles can rotate. Twenty-five minutes on a real task, then a 60-second report: our task, where AI could help, what a human must still do, the main risk, how we would check it.

1

What is the task?

2

How do we currently do it?

3

Could AI help?

4

Which part could AI do?

5

Which part needs a human?

6

What could go wrong?

7

What would we need to check?

The last two questions are the ones that matter, and the ones groups skip if nobody pushes.

Help Aoife decide where AI could genuinely help.

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What did we learn?

Not another lecture — the facilitator asks, the room answers, 3–5 observations go on the board.

1. Where did AI look genuinely useful?
2. Where did we decide NOT to use AI?
3. What surprised you?
4. What could go wrong?
5. What information would you not give an AI system?
6. Who remains responsible for the final result?

MORNING IDEA

AI can help, but it has limitations.

LESSON

“Can AI do it?” is not the same question as “Should AI do it?”

AI is an assistant, not an authority.

The prompt challenge

Everyone starts from the same weak prompt. Four rungs. If a detail is missing, leave a clearly marked gap rather than inventing it, and list any Irish terms you were unsure about.

Prompt 1

Write a newsletter about our business.

Prompt 2

+ who it is for, and what it is for

Audience

Purpose

Prompt 3

+ how it should sound, how long, and the actual facts

Tone

Length

Key information

Prompt 4

+ shape, limits, source discipline and language

Format

Avoid

Source

Irish + English

Prompt engineering is communication — a better prompt makes the answer more useful, not more true.

Aoife's first prompt isn't very good. Improve it, rung by rung.

Coffee break

14:40 – 14:55 · Fifteen minutes

After the break: what changed when we improved the prompt?

What changed?

Ten minutes. Same method as Reflection 1: ask, capture, connect to the morning.

1. What changed when we improved the prompt?
2. Which instruction made the biggest difference?
3. Did the better prompt eliminate all errors?
4. What did the AI still misunderstand?
5. What would you check before using the answer?
6. Would you use the same prompt again?

PREDICTION

The model responds to the information we give it.

FLUENCY

A polished answer can still contain errors.

IRISH

A fluent-looking Irish answer still needs human review.

A better prompt can improve an answer. It does not remove the need for judgement.

STOP

Four steps, in order, between reading an AI output and doing anything with it.

S

Stop

Do not immediately trust or publish.

The gap between reading and acting is where all the safety lives.

T

Think

What exactly is being claimed?

Separate the claims from the prose. A paragraph can hold four assertions.

O

Observe

What evidence do we have?

“The model said so” is not evidence. Ask it to quote, or check yourself.

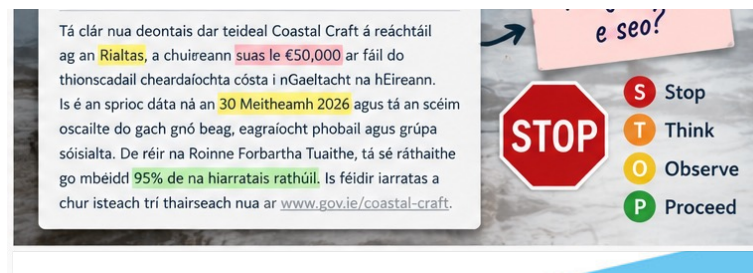
P

Proceed

Only after checking.

And with a named person attached. The tool never carries the consequence.

AI is an assistant, not an authority.



My first AI experiment

Fifteen minutes, written alone, no discussion. This is what connects today to Session 2.

One real task I want to try

Something from next week, not a test.

AI tool I might use

What I expect it to help with

Time, a first draft, ideas, structure.

What could go wrong

Be specific.

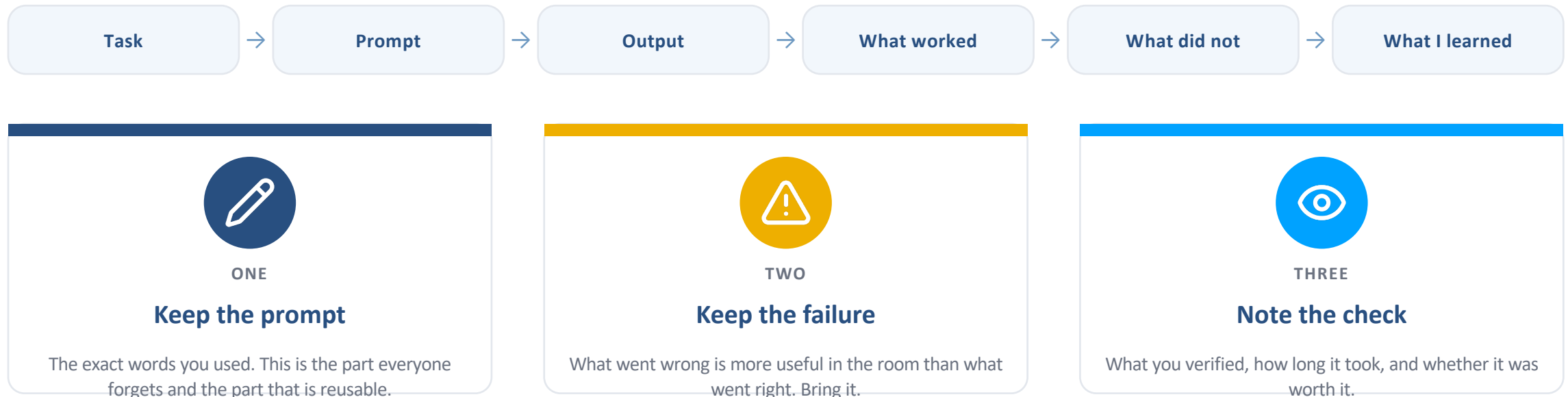
How I will check the result

The method, and who does it.

Also note: what information will I NOT put into the AI, and who makes the final decision? Bring the one that did not work — it teaches the room more than the one that did.

Try AI three times

Three different real work tasks. Not exercises. Record each one and bring the record with you.



Also note how long the task took, how long checking took, and whether it was worth using. Bring a failure — it teaches the room more than a success. Failures are the valuable material.

What we said today

Aoife's day is nearly finished. This morning she asked why she doesn't use AI. What would she do differently tomorrow?

You do not need to become an AI expert. You need to become confident enough to make good decisions about AI.

AI is an assistant, not an authority.

Prediction is not understanding.

Fluent is not the same as factual.

You remain responsible for what you use and publish.

AI for Irish — not AI instead of Irish.

What happens next

Responsible AI and AI for everyday business. Built partly from what you bring back.



FIRST HOUR

Your three experiments

What worked, what failed, what surprised you. This shapes the rest of the day.



MORNING

The AI Act, in plain language

What it actually asks of a small organisation. Educational guidance, not legal advice.



MORNING

Safety with teeth

Privacy, confidentiality, what never goes in, and the STOP check for inputs — plus the deeper AI-judges-AI and trust/verify exercises we made room for today.



AFTERNOON

AI for everyday business

Communication, documents, research, and turning one prompt into a repeatable workflow.

Everything on the concerns wall gets an answer, or an honest “not yet”.



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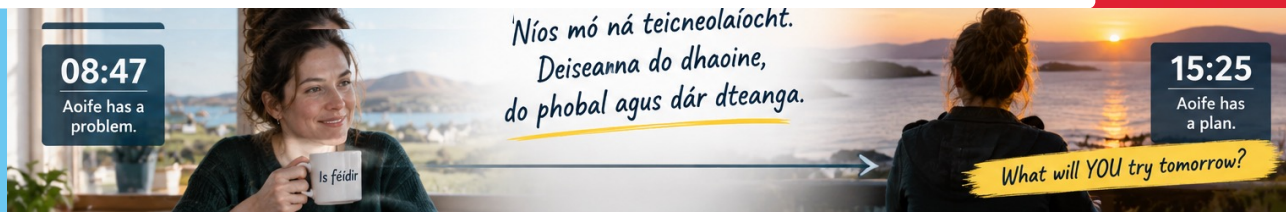
Go raibh maith agaibh

AI is an assistant, not an authority.

AI for Irish — not AI instead of Irish.

Dr Haithem Afli

Human-Centred AI | NLP | Machine Translation | Responsible AI



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